

Introduction: DNA carries the fundamental information of living beings, encoding their biological structure and function. However, this concept extends beyond Earth—galaxies across the universe have their own "chemical DNA" expressed in emission lines. The light emitted by galaxies spreads out to form a spectrum, which acts as a galaxy's unique fingerprint and contains specific markers called emission lines. Captured by the most sophisticated telescopes, these lines carry information such as the galaxy's gas density, and how much hydrogen, oxygen, and other elements are present in the galaxy (1). The study of these lines is called spectroscopy. However, these lines get fainter with distance, and spectroscopic studies of them become challenging. To overcome this problem, astronomers use a technique called spectral stacking, which combines the spectra of multiple galaxies to enhance weak signals. Still, whether stacks have been used to reliably infer the characteristics of individual galaxies has not been thoroughly analyzed. My research aims to assess whether the stacking technique accurately represents the properties of individual galaxies, ensuring its reliability for studying distant galaxies.

Background: As galaxies move farther away from us, they are also moving the fastest away from us, and their light shifts to longer (redder) wavelengths—a phenomenon called redshift. Projects like the James Webb Space Telescope (JWST) and the Keck Baryonic Structure Survey (KBSS)—a spectroscopic survey designed to study the gas surrounding high-redshift galaxies—are set to improve our understanding of galactic formation and evolution at early cosmic times (2). Despite their potential to gather spectra from galaxies over 10 billion light-years away, the properties from their emission lines do not always come easily. Traditionally, one approach is looking at galaxies individually, but this is not always possible and effective. This is because the emission lines are too faint and become even fainter the further in space we look at the galaxy.

However, there is a way of amplifying and getting clearer emission lines: the technique of spectral 'stacking'. This method combines multiple galactic spectra to enhance signal strength, which has been heavily used by researchers to study faint emission lines (3). Stacking spectra layers the fluxes of similar galaxies into one improved spectrum, enhancing the signal-to-noise ratio (S/N). Yet, the relationship between individual galaxy properties and those measured from stacked combinations of galaxy spectra is unclear, demonstrating the demand for research in galactic stacking.

This project builds on my SURG research, where I evidenced how spectral stacking can be used to amplify faint emission lines from high-redshift galaxies. During the SURG, I worked with spectra from a sample of 26 galaxies collected using the KBSS and 24 spectra from the Chemical Evolution Constrained using Ionized Lines in Interstellar Aurorae (CECILIA) Survey, which relies on data from NASA's JWST. Although my work showcased the potential of stacking to reveal emission lines, the resulting stacks still require further evaluation. Specifically, I want to analyze how well stacks preserve key spectral features from single galaxies, such as elemental abundances (e.g., the amount of oxygen present) and how dense the galactic gas is.

Therefore, by refining the stacking technique, my project will enhance our understanding of distant galaxies and contribute to astrophysics and data analysis, paving the way for new insights into galaxy formation and evolution. **How reliable is the stacking process as a tool to study distant galaxies?**

Methodology: To compare the measurement from individual and stacked spectra, I will look at two specific properties: "metallicity" and gas density. 'Metals' in astronomy refer to elements heavier than hydrogen and helium in a galaxy and metallicity is the abundance of metal in the galaxy. The data I will work with are the spectra for both single galaxies and stacks

produced during my first SURG (Fig 1.). These plots were produced by refining the raw data with corrections. First, I applied redshift corrections, aligning each galaxy's spectrum to a common rest-frame wavelength for comparison of their emission lines. Then, I performed slit-loss corrections, which account for instrumental effects that distort measured fluxes for each galaxy. Thereafter, I executed wavelength interpolation, placing all spectra onto a uniform wavelength grid, necessary for combining fluxes during stacking. To minimize noise, I utilized sigma clipping, removing extraneous data points that were not representative of the sample. With the corrected data, I plotted the stacks based on the sum, mean, median, and standard deviation of the fluxes.

Using the stacks with a better S/N ratio, I will first analyze the metallicity of the galaxy. The metallicity will be measured by the “strong line method.” (4) I first need to locate strong (often the brightest and easiest to detect) emission lines in the spectrum by identifying the wavelength positions of the peaks obtained. Each element emits light at specific wavelengths, already known and well-documented through atomic physics studies. Thus, I can compare the peaks in the data to the known wavelengths of elements such as oxygen and nitrogen. Next, I will determine the flux of each element by calculating the area under the peak, and compute the line ratio by normalizing the fluxes relative to a reference line (usually hydrogen). These ratios are then plugged into empirical mathematical relationships that compare observed ratios to known metallicities from other galaxies or models, thus determining the galaxy's metal content.

I will measure gas density by comparing the ratios of specific emission lines in both the individual spectra and the stack. I will do this by analyzing the brightness of two emission lines that are close in wavelength but respond differently to how dense the gas is, because of atomic physics (5). By comparing the line ratio measurements from individual galaxies to those from the stack, I will see how they differ. This analysis will reveal how well the stack represents the typical properties of the sample—whether it reflects the mean, median, or another value—and quantify any discrepancies, exploring potential inaccuracies caused by the stacking technique.

Qualifications: My academic and research experiences have equipped me with the necessary skills to successfully execute this research. Through my SURG, I worked with spectral data from the KBSS and the CECILIA survey, where I applied the key processing techniques and coding skills I will use in this project. Additionally, courses like Programming for Data Science, Mathematical Tools for Physics, and Highlights of Astronomy provided me with primary knowledge in programming, data analysis, and galaxy evolution, preparing me for both the theoretical and technical parts of this research. Furthermore, my participation in an observing run at the Keck Observatory in Hawaii provided hands-on experience with raw spectral data collection and analysis. Ultimately, this project will prepare me for future research in extragalactic astronomy, an area in which I am interested in pursuing graduate studies. This opportunity will strengthen both my theoretical and computational skills, laying the foundation for my career as a scientific researcher in Astrophysics.

Citations (MLA)

- (1) Izotov, Yuri I., et al. "The chemical composition of metal-poor emission-line galaxies in the Data Release 3 of the Sloan Digital Sky Survey. " *Astronomy & Astrophysics* 448.3 (2006): 955-970.
- (2) Nagamine, Kentaro, et al. "Probing Feedback via IGM tomography and the Ly α Forest with Subaru PFS, TMT/ELT , and JWST . " *The Astrophysical Journal* 914.1 (2021): 66.
- (3) Strom, Allison L., et al. "CECILIA: The Faint Emission Line Spectrum of $z \sim 2-3$ Star-forming Galaxies. " *The Astrophysical Journal Letters* 958.1 (2023): L11. of Spatially Extended Ly α Emission in the KBSS-KCWI Survey"(2023, ApJ, 953, 118). " *The Astrophysical Journal* 961.1 (2024): 143.
- (4) Maiolino, Roberto, and Fillipo Mannucci. "De re metallica: the cosmic chemical evolution of galaxies." *The Astronomy and Astrophysics Review* 27.1 (2019): 3.
- (5) Sparke, Linda S., and John Sill Gallagher. *Galaxies in the universe: an introduction*. Cambridge University Press, 2000.

Appendix A: Figures

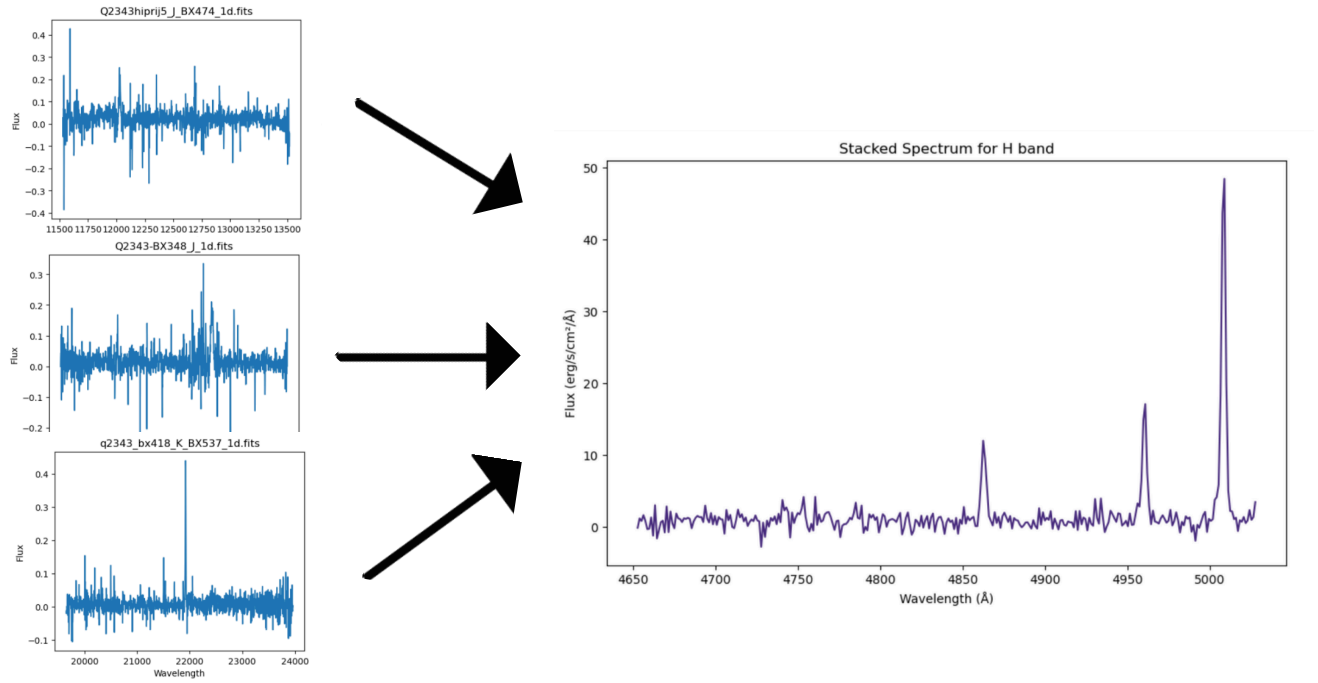


Figure 1. Individual spectra for and Stacked spectrum for galaxies in H Band

The left graphs show individual galaxy spectra for galaxies in the H-band (wavelengths in the infrared range). The right graph represents the stacked spectrum in the H-band after combining the multiple individual graphs, enhancing the visibility of the emission lines.